



IPv4 Subnetting & VLSM

Use this when: you size a subnet, find the network, broadcast and host range, or plan addresses with VLSM.

CCNA 200-301 · topic 1.6
Sheet v1.0 · Sep 2026

Scan to visit

Subnet chart

Hosts = $2^{\text{host bits}} - 2$

BIT VALUE	128	64	32	16	8	4	2	1	
MASK OCTET	128	192	224	240	248	252	254	255	
CIDR	MASK	WILDCARD	BLOCK	HOSTS	CIDR	MASK	WILDCARD	BLOCK	HOSTS
/24	255.255.255.0	0.0.0.255	256	254	/16	255.255.0.0	0.0.255.255	65,536	65,534
/25	...255.128	0.0.0.127	128	126	/17	...128.0	0.0.127.255	32,768	32,766
/26	...255.192	0.0.0.63	64	62	/18	...192.0	0.0.63.255	16,384	16,382
/27	...255.224	0.0.0.31	32	30	/19	...224.0	0.0.31.255	8,192	8,190
/28	...255.240	0.0.0.15	16	14	/20	...240.0	0.0.15.255	4,096	4,094
/29	...255.248	0.0.0.7	8	6	/21	...248.0	0.0.7.255	2,048	2,046
/30	...255.252	0.0.0.3	4	2	/22	...252.0	0.0.3.255	1,024	1,022
/31	...255.254	0.0.0.1	2	2*	/23	...254.0	0.0.1.255	512	510
/32	...255.255	0.0.0.0	1	1	/8	255.0.0.0	0.255.255.255	16.7 M	16.7 M

Solve any IP in 4 steps

Given 192.0.2.77/27

- Mask for /27 **255.255.255.224**
- Block size = 256 - 224 **32**
- Network = multiple of 32 ≤ 77 **192.0.2.64**
- Broadcast = next network - 1 **192.0.2.95**

Usable hosts: .65 - .94 (30). Gateway is usually the first: .65

VLSM: carve 192.0.2.0/24

Largest first

	LAN A /25	LAN B /26	C /27	free
LAN A · 100 hosts	.0/25	.1 - .126		
LAN B · 50 hosts	.128/26	.129 - .190		
LAN C · 20 hosts	.192/27	.193 - .222		
WAN 1 · 2 hosts	.224/30	.225 - .226		
WAN 2 · 2 hosts	.228/30	.229 - .230		

Special ranges

RFC 6890

10.0.0.0/8 · 172.16.0.0/12 · 192.168.0.0/16	Private (RFC 1918)
127.0.0.0/8	Loopback: this host
169.254.0.0/16	APIPA: DHCP failed
100.64.0.0/10	Carrier-grade NAT
224.0.0.0/4	Multicast
0.0.0.0/8 · 255.255.255.255	"This network" · limited broadcast
192.0.2.0/24 · 198.51.100.0/24 · 203.0.113.0/24	Documentation only

Check & set addresses

ipconfig · ip -4 addr	Win · Linux
show ip interface brief	Cisco
show ip route connected	Cisco subnets
route print -4 · ip route	Win · Linux
ip address 192.0.2.65 255.255.255.224	

CAUTION

CAUTION Re-addressing the interface you are connected through cuts your SSH session. Use the console or a second path.



REMEMBER

- Mask octets: **128 192 224 240 248 252 254 255**
- Network and broadcast are not usable hosts
- A PC with **169.254.x.x** never got a DHCP lease
- Wildcard = **255.255.255.255 - mask** (ACLs, OSPF)
- * **/31** = 2 usable on point-to-point links (RFC 3021)
- VLSM: sort by size, **largest subnet first**